

EPPS 6354 Information Management

Assignment 2

Answer all questions.

Prepare your answers in presentation format and be ready to present in class

1. What are the differences between *relation schema*, *relation* and *instance*? Give an example using the university database to illustrate.
2. Draw a schema diagram for the following bank database:

```
branch(branch_name, branch_city, assets)
customer (ID, customer_name, customer_street, customer_city)
loan (loan_number, branch_name, amount)
borrower (ID, loan_number)
account (account_number, branch_name, balance)
depositor (ID, account_number)
```

3. Consider the above bank database. Assume that branch names (*branch_name*) and customer names (*customer_name*) uniquely identify branches and customers, but loans and accounts can be associated with more than one customer.
 - i. What are the appropriate primary keys? (Underline each in diagram)
 - ii. Given your choice of primary keys, identify appropriate foreign keys.
4. (Post on website) Describe two ways artificial intelligence or LLM can assist in managing or querying a database. In your answer, briefly explain how each method improves efficiency or accuracy compared to traditional (non-AI) approaches. (3–5 sentences)